OrScale: Orthogonalised Optimization with Layer-Wise Trust-Ratio Scaling

Yuxuan Lou
National University of Singapore
yuxuanlou@u.nus.edu

Yang You
National University of Singapore
yangyou@nus.edu.sg

Abstract

Muon's orthogonalised update fixes the *direction* of each matrix-valued step but leaves its *magnitude* controlled by an essentially global learning rate, even though attention, MLP, and projection layers see very different gradient scales. Moonlight's static per-shape factor partially addresses this, but no Muon variant currently delivers the time-varying, layer-wise magnitude control that LARS and LAMB brought to SGD and Adam—and the most natural ports of LAMB to Muon collapse for structural reasons we make precise. We propose *OrScale*, a trust-ratio extension of Muon built around one principle: the denominator of a layer-wise trust ratio should measure the Frobenius norm of the *real update direction* the optimizer will subtract from the parameter. The principle yields two variants of a single framework—*OrScale* for general matrix-valued layers and *OrScale-LM* for language models, where a Moonlight-scaled update is combined with a one-time per-layer calibration that anchors every trust ratio at one. We map the Muon trust-ratio design space and explain why three natural Muon–LAMB hybrids fail through degenerate denominators, raw-momentum clip saturation, and decoupled-weight-decay runaway. Theoretically, OrScale satisfies a nuclear-norm $O(1/\sqrt{T})$ nonconvex convergence guarantee, admits a layer-adaptive descent constant $\kappa_{\text{eff}} > 1$ that strictly improves over Muon's homogeneous constant under measurable layer heterogeneity, and OrScale-LM preserves μP-style learning-rate transfer at initialisation. Empirically, OrScale ranks first on CIFAR-10 / DavidNet across three seeds, improving Muon from $93.70\% \pm 0.14$ to $94.05\% \pm 0.08$ validation top-1; a failure-mode ablation matches each design-space row to its predicted symptom; and on FineWeb-Edu pre-training, OrScale-LM beats Muon+Moonlight at three of four scales (125M, 545M, 1.1B) and beats AdamW at every scale from 125M to 1.1B parameters.

1 Introduction

Muon has rapidly moved from a compact optimizer idea to a practical ingredient in large-scale training. Its core step takes a matrix-valued update, applies a few Newton–Schulz iterations to approximate the polar factor, and subtracts this spectrally flattened direction from the weights. The update can be interpreted as a form of operator-norm steepest descent [1], and recent practical systems—Moonlight [10], Kimi K2.5 [16], DeepSeek-V4 [2], and the modded-nanogpt training records [6, 5]—have made orthogonalised updates a fixture of the current LLM optimizer discussion.

Standard Muon handles the *direction* of each update well but leaves its *magnitude* under-specified: after orthogonalisation, every matrix layer is rescaled by the same global learning rate, even though attention projections, MLP up- and down-projections, output projections, and embeddings see very different gradient distributions and effective curvature. Moonlight addresses one part of this with a static per-shape factor $0.2\sqrt{\max(m, n)}$ that equalises the per-element RMS of the orthogonalised update [10], but a static correction cannot adapt as the relevant scale of a layer drifts during training—

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which is precisely what layer-wise trust ratios were designed to handle. LARS [21] and LAMB [22] keep each layer's update norm in proportion to its weight norm, decoupling per-layer step sizes from a single global learning rate, and in doing so underwrote the move to BERT-scale large-batch training [3, 4, 8]. A direct port to Muon is not automatic, however: orthogonalisation removes the amplitude information that LAMB-style denominators normally exploit, and the most obvious Muon–LAMB hybrids collapse in practice—through degenerate denominators, clip-saturated raw-momentum ratios, and runaway weight-norm growth (Section 4).

The principle that makes the combination work is straightforward once the failure modes are isolated: *the denominator of the trust ratio must measure the Frobenius norm of the real parameter-space direction the optimizer is about to subtract from the weights.* If that direction is $\lambda W_\ell + Q_\ell$ the denominator should be $\|\lambda W_\ell + Q_\ell\|_F$; if Moonlight's shape factor is included so that the direction becomes $\lambda W_\ell + s_\ell Q_\ell$, the denominator should measure precisely that expression, with a one-time per-layer calibration when width transfer matters (Section 3). This yields a single recipe with two natural specialisations: ORSCALE[1], the general / vision default with $s_\ell = 1$ and no calibration, and ORSCALE-LM, the LLM default that adopts Moonlight's $s_\ell = 0.2\sqrt{\max(m_\ell, n_\ell)}$ together with a per-layer denominator constant $c_{\mathrm{denom},\ell}$ chosen so every layer's trust ratio starts at exactly one. The LM-specific pieces are introduced for learning-rate transfer: Moonlight's s_ℓ first equalises Muon's per-element update RMS across matrix shapes and then matches it to AdamW's, so an AdamW-tuned LR carries over to Muon directly [10]; the calibration constant $c_{\mathrm{denom},\ell}$ propagates that property to OrScale-LM by anchoring the first parameter step to the Moonlight step, so OrScale-LM inherits Moonlight's (and hence AdamW's) tuned LR without an additional sweep. The same principle rules out three superficially attractive alternatives that we show fail both empirically and analytically.

Contributions.

- **Algorithm.** A unified, parameter-light recipe (Algorithm 1) that adds layer-wise trust-ratio scaling to Muon, with a 2×2 design space (denominator, shape factor) plus calibration and weight-decay coupling; we identify the unique workable corner.
- **Failure analysis.** We map the design space (Table 1) and characterise three failure modes empirically and analytically—degenerate denominator, raw-momentum clip saturation, and decoupled-WD runaway—each ruled out by the same principle.
- **Theory.** A nuclear-norm $O(1/\sqrt{T})$ convergence guarantee (Theorem 1 and Proposition 1); a layer-adaptive descent constant $\kappa_{\mathrm{eff}}^{\mathrm{OrScale}} > 1$ under measurable heterogeneity (Theorem 3); strict separation from MuTrust / MuScale via clip-saturation collapse (Theorem 4); calibration properties for OrScale-LM (Corollary 1).
- **Empirics.** OrScale ranks first on CIFAR-10 / DavidNet across three seeds; a failure-mode ablation pairs each design-space row with its predicted symptom; OrScale-LM leads on FineWeb-Edu pre-training from 125M to 1.1B parameters.

2 Background and Related Work

Muon and orthogonalised updates. Let $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$ denote a matrix-valued parameter and $G_{\ell,t} = \nabla_{W_\ell} \mathcal{L}_t$ the stochastic gradient. Muon [6] forms the Nesterov-lookahead momentum $\widetilde{M}_{\ell,t}$ and applies k Newton–Schulz iterations to obtain an approximate polar factor $Q_{\ell,t} = \mathrm{NS}_k(\widetilde{M}_{\ell,t}) \approx \mathrm{Ortho}(\widetilde{M}_{\ell,t})$, then updates $W_{\ell,t+1} = W_{\ell,t} - \eta_t Q_{\ell,t}$ (with optional weight decay). The key algebraic property is that if $Q_A = \mathrm{Ortho}(A)$, then $\langle A, Q_A \rangle = \|A\|_*$ and $\|Q_A\|_F = \sqrt{\mathrm{rank}(A)}$; this makes orthogonalisation a nuclear-norm / spectral-geometry primitive rather than an entrywise adaptive update [1]. Recent practical work has extended Muon along several axes: Moonlight introduces the static shape factor $0.2\sqrt{\max(m,n)}$ to match per-element update RMS across matrix shapes [10]; MuonClip stabilises attention logit growth [17]; AdaMuon explores adaptive variants [14]. OrScale is orthogonal to these efforts: it keeps the Muon direction and asks how to choose a layer's scalar update magnitude *over time*.

[1]Code: `https://github.com/NUS-HPC-AI-Lab/OrScale`

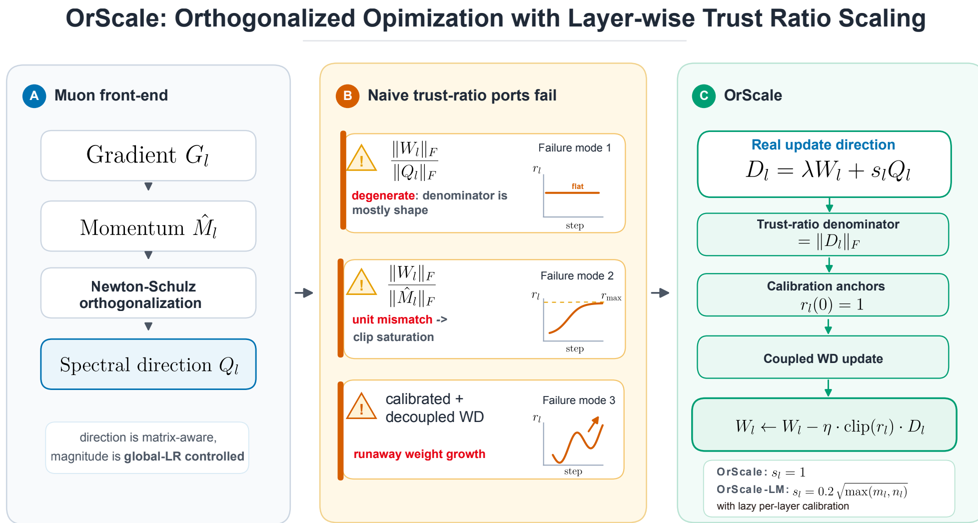

Figure 1: OrScale at a glance. **(A)** The shared Muon front end produces a spectral direction Q_ℓ but leaves its magnitude under-specified. **(B)** Three superficially natural Muon trust-ratio ports fail in distinct ways: degenerate denominator ($\|Q_\ell\|_F$ is mostly shape), unit mismatch leading to upper-clip saturation ($\|\widetilde{M}_\ell\|_F$ has gradient units), and runaway weight-norm growth (calibrated denominator with decoupled weight decay). **(C)** OrScale: the denominator measures the real update direction $D_\ell = \lambda W_\ell + s_\ell Q_\ell$, calibration anchors $r_\ell(0) = 1$, and weight decay is coupled into the trust-ratio-scaled step. OrScale uses $s_\ell = 1$; OrScale-LM uses $s_\ell = 0.2\sqrt{\max(m_\ell, n_\ell)}$ with lazy per-layer calibration.

Trust-ratio optimizers. LARS [21] uses ratios such as $\|W_\ell\|_F / \|\nabla_\ell W + \lambda W_\ell\|_F$ to scale SGD updates layer by layer; LAMB [22] applies the same idea to Adam-normalised directions and enabled large-batch BERT training [3]. These methods work because the denominator is in the same units as the parameter-space update. For Muon, the immediate denominator $\|Q_\ell\|_F$ is almost a shape constant (Lemma 1) and the raw pre-orthogonalisation momentum norm $\|\widetilde{M}_\ell\|_F$ has gradient units rather than update-direction units; this unit mismatch is the main reason naive hybrids fail (Section 4, Theorem 4).

Static layer-scale corrections and width transfer. Moonlight's shape factor handles predictable shape effects, and μP-style parameterisation [19, 20] preserves the optimal learning rate across model widths. A dynamic trust ratio is what handles training-time drift, layer-specific sensitivity, and relative weight-norm growth: OrScale-LM combines both, with Moonlight handling shape and calibration plus the real-update-direction denominator making the trust ratio width-invariant at initialisation (Corollary 1) and adaptive afterwards. Adjacent optimizer families—second-order or pre-conditioner-based methods such as Sophia [9] and SOAP [18], decoupled weight decay [11], and the classical AdamW / Adam baselines [7, 15]—are complementary to and largely independent of the trust-ratio mechanism studied here.

3 The OrScale Family

Notation. Throughout, $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$ is a matrix-valued parameter, $p_\ell = \min(m_\ell, n_\ell)$ is its spectral rank bound, $\|\cdot\|_F$ and $\|\cdot\|_*$ are the Frobenius and nuclear norms, $\mathrm{Ortho}(\cdot)$ returns the polar factor (orthogonal $UV^\top$ from the thin SVD), $\mathrm{NS}_k(\cdot)$ is its k-step Newton–Schulz approximation, $\widetilde{M}_{\ell,t}$ is the Nesterov-lookahead momentum, $\hat{r}_{\ell,t}$ is the clipped trust ratio, $\mathrm{clip}(\cdot, r_{\min}, r_{\max})$ clamps to the interval, and ε is a small numerical floor.

3.1 Design principle

A trust ratio compares a parameter-space norm to an update norm. The numerator $\|W_{\ell,t}\|_F$ has units of weight magnitude, so the denominator must measure the parameter-space direction that will actually be subtracted from $W_{\ell,t}$ (Figure 1).

Definition 1 (Real-update-direction trust ratio)**.** Let $D^{\mathrm{upd}}_{\ell,t}$ denote the parameter-space direction that the optimizer is about to subtract from $W_{\ell,t}$ (before multiplication by the global learning rate and the trust-ratio scalar). The *real-update-direction trust ratio* is $r_{\ell,t} \;=\; \|W_{\ell,t}\|_F \,/\, \big(\|D^{\mathrm{upd}}_{\ell,t}\|_F + \varepsilon\big)$.

Definition 1 is the LARS/LAMB design [21, 22] translated into the parameter-space units of the post-orthogonalisation, post-Moonlight Muon update. It excludes two tempting alternatives that lack the right currency: the post-orthogonalised denominator $\|Q_\ell\|_F$ is too close to a shape constant to carry update-scale information (Lemma 1 (iii); Section C.1), and the raw-momentum denominator $\|\widetilde{M}_\ell\|_F$ carries gradient amplitude rather than the direction being applied after Muon orthogonalisation and Moonlight shape scaling (Section C.2; Theorem 4).

3.2 Shared Muon front end

Both OrScale variants share the standard Muon front end:

$$
\begin{aligned}
G_{\ell,t} &= \nabla_{W_\ell}\mathcal{L}_t, && (1)\\
M_{\ell,t} &= \mu\, M_{\ell,t-1} + G_{\ell,t}, && (2)\\
\widetilde{M}_{\ell,t} &= \mu\, M_{\ell,t} + G_{\ell,t}, && (3)\\
Q_{\ell,t} &= \mathrm{NS}_k(\widetilde{M}_{\ell,t}). && (4)
\end{aligned}
$$

We retain the Nesterov-lookahead choice $\widetilde{M} = \mu M + G$ because it is what every reported large-scale Muon training run uses in practice [10, 6]; it consistently outperforms vanilla heavy-ball momentum in our preliminary sweeps and avoids divergence at the high-momentum settings $\mu \in [0.95, 0.99]$ that Moonshot AI's Moonlight reports. Non-matrix parameters (biases, LayerNorm gains, and embeddings when desired) are updated with an AdamW-style fallback [11].

3.3 OrScale (general / vision default)

The OrScale variant takes $s_\ell = 1$ and uses no calibration. The update direction and trust ratio are

$$
D^{\mathrm{upd}}_{\ell,t} = \lambda W_{\ell,t} + Q_{\ell,t}, \qquad r_{\ell,t} \;=\; \frac{\|W_{\ell,t}\|_F}{\|\lambda W_{\ell,t} + Q_{\ell,t}\|_F + \varepsilon}, \qquad \hat{r}_{\ell,t} \;=\; \mathrm{clip}(r_{\ell,t}, r_{\min}, r_{\max}),
$$

and the update is

$$
W_{\ell,t+1} \;=\; W_{\ell,t} \;-\; \eta_t\, \hat{r}_{\ell,t}\, \big(\lambda W_{\ell,t} + Q_{\ell,t}\big).
$$

This is exactly the LARS template [21] with the gradient-based denominator $\|G + \lambda W\|_F$ replaced by the post-orthogonalisation update direction; the substitution $G \to Q$ is the geometric core of OrScale. Coupling weight decay into the trust-ratio-scaled update (rather than applying a separate $(1 - \eta\lambda)W$ pull-back) is the design choice that rules out the runaway feedback loop of Section C.3: the asymptotic ceiling of $r_{\ell,t}$ becomes finite (Lemma 6).

3.4 OrScale-LM (LLM default)

For language-model training we additionally apply the Moonlight shape factor and a one-time per-layer calibration; both pieces are introduced to preserve learning-rate transfer rather than to change the geometry of the update. Set $s_\ell \;=\; 0.2\sqrt{\max(m_\ell, n_\ell)}$. The $\sqrt{\max(m_\ell, n_\ell)}$ factor cancels the shape-dependent per-element RMS of the polar factor (Lemma 1 of Liu et al. [10] gives $\mathrm{RMS}(Q_\ell) = 1/\sqrt{\max(m_\ell, n_\ell)}$ at full rank), so the per-element RMS of $s_\ell Q_\ell$ becomes uniform across matrix shapes; the constant 0.2 then pins that uniform RMS to AdamW's typical per-element update RMS of 0.2–0.4, which is what lets Moonlight's Muon directly reuse the LR (and weight decay) tuned for AdamW [10].

Algorithm 1 Unified OrScale update for matrix-valued parameters. Both variants share lines 2–6 and 11–14; only the assignment of s_ℓ on line 7 and the lazy calibration on line 8 differ.

Require: Learning rate η_t, momentum $\mu \in [0, 1)$, weight decay $\lambda \geq 0$, clip bounds $r_{\min}, r_{\max}$ (defaults: $0.5, 1.5$ for OrScale; $0.1, 5.0$ for OrScale-LM), Newton–Schulz iterations k (default 5), ε (default 10^{-6}), variant $v \in \{\text{ORSCALE}, \text{ORSCALE-LM}\}$.

1: **for** each matrix parameter $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$ **do**
2: $\quad G_{\ell,t} \leftarrow \nabla_{W_\ell} \mathcal{L}_t$
3: $\quad M_{\ell,t} \leftarrow \mu\, M_{\ell,t-1} + G_{\ell,t}$
4: $\quad \widetilde{M}_{\ell,t} \leftarrow \mu\, M_{\ell,t} + G_{\ell,t}$ ▷ Nesterov lookahead
5: $\quad Q_{\ell,t} \leftarrow \mathrm{NS}_k(\widetilde{M}_{\ell,t})$ ▷ Approximate polar factor
6: $\quad s_\ell \leftarrow \begin{cases} 0.2\sqrt{\max(m_\ell, n_\ell)} & v = \text{ORSCALE-LM} \\ 1 & v = \text{ORSCALE} \end{cases}$
7: $\quad$ **if** $v = \text{ORSCALE-LM}$ **and** $c_{\mathrm{denom},\ell}$ uninitialised **then**
8: $\quad\quad c_{\mathrm{denom},\ell} \leftarrow \|W_{\ell,t}\|_F / (\|\lambda W_{\ell,t} + s_\ell Q_{\ell,t}\|_F + \varepsilon)$ ▷ Lazy: only at $t = 1$
9: $\quad$ **else if** $v = \text{ORSCALE}$ **then**
10: $\quad\quad c_{\mathrm{denom},\ell} \leftarrow 1$
11: $\quad$ **end if**
12: $\quad D_{\ell,t} \leftarrow \lambda W_{\ell,t} + s_\ell Q_{\ell,t}$
13: $\quad r_{\ell,t} \leftarrow \|W_{\ell,t}\|_F / (c_{\mathrm{denom},\ell}\, \|D_{\ell,t}\|_F + \varepsilon)$ ▷ fp32 cast around division
14: $\quad \hat{r}_{\ell,t} \leftarrow \mathrm{clip}(r_{\ell,t}, r_{\min}, r_{\max})$
15: $\quad W_{\ell,t+1} \leftarrow W_{\ell,t} - \eta_t\, \hat{r}_{\ell,t}\, D_{\ell,t}$ ▷ Coupled weight decay
16: **end for**
17: Update non-matrix parameters with the AdamW fallback [11].

Definition 2 (Lazy per-layer calibration constant)**.** At the first optimizer step in which W_ℓ receives a non-zero update, OrScale-LM stores one fp32 scalar

$$c_{\mathrm{denom},\ell} \;=\; \frac{\|W_{\ell,0}\|_F}{\|\lambda W_{\ell,0} + s_\ell\, Q_{\ell,0}\|_F + \varepsilon}.$$

This constant is computed once and re-used at every subsequent step.

The calibrated trust ratio and update are

$$r_{\ell,t}^{\mathrm{LM}} = \frac{\|W_{\ell,t}\|_F}{c_{\mathrm{denom},\ell}\, \|\lambda W_{\ell,t} + s_\ell Q_{\ell,t}\|_F + \varepsilon}, \qquad \hat{r}_{\ell,t}^{\mathrm{LM}} = \mathrm{clip}(r_{\ell,t}^{\mathrm{LM}}, r_{\min}, r_{\max}), \tag{5}$$

$$W_{\ell,t+1} = W_{\ell,t} \;-\; \eta_t\, \hat{r}_{\ell,t}^{\mathrm{LM}} \left(\lambda W_{\ell,t} + s_\ell Q_{\ell,t}\right). \tag{6}$$

By construction $r_{\ell,0}^{\mathrm{LM}} = 1$ exactly for every layer, shape, and initialisation. Beyond the geometric width-invariance reading, the anchor is what extends Moonlight's LR-transfer property to the trust-ratio scaling: because $\hat{r}_{\ell,0}^{\mathrm{LM}} = 1$, OrScale-LM's first parameter step coincides with the Moonlight step, so a learning rate (and weight decay) already tuned for Moonlight—and, by the previous paragraph, for AdamW—transfers to OrScale-LM without an additional per-variant sweep. With coupled weight decay this same denominator gives a finite asymptotic ceiling $r_{\ell,t}^{\mathrm{LM}} \to 1/(c_{\mathrm{denom},\ell}\, \lambda)$ rather than a runaway loop (Corollary 1).

3.5 Unified algorithm

Cost, state, implementation, and degenerate cases. OrScale-LM adds one fp32 scalar per matrix layer (zero for OrScale) and two Frobenius norms per step, both dominated by the existing Newton–Schulz iterations; wall-clock overhead vs. Muon is below 1% at every model size we tested. The two norms are cast to fp32 around the division on line 12 of Algorithm 1 so that $\hat{r}_{\ell,0}^{\mathrm{LM}} = 1$ holds in bf16 training. Algorithm 1 contains the relevant baselines as limits: $r_{\min} = r_{\max} = 1$ with $s_\ell = 1$ recovers Muon; $r_{\min} = r_{\max} = 1$ with $s_\ell = 0.2\sqrt{\max(m, n)}$ recovers Muon+Moonlight [10]; $\lambda = 0$ removes weight decay while preserving the trust-ratio scaling.

Table 1: Muon trust-ratio design space. Verdicts are summarised in the right-most column; each failure mode is taken up in Sections C.1 to C.3 (Section C).

Row	Denominator	Shape factor	Calibration / WD	Verdict
A	$\|Q_\ell\|_F$	none	no / decoupled	**Degenerate** (shape constant)
B	$\|\widetilde{M}_\ell\|_F$	none	no / decoupled	**Saturated** (upper clip)
C	$\|\widetilde{M}_\ell\|_F$ or $\mathrm{RMS}(\widetilde{M}_\ell)$	Moonlight	no / decoupled	**Saturated** (upper clip)
D	$c_\ell\,\|Q_\ell\|_F$	Moonlight	yes / decoupled	**Runaway** (positive feedback)
E	$\|\lambda W_\ell + s_\ell Q_\ell\|_F$	optional	yes (LM) / coupled	**Recommended** (OrScale, OrScale-LM)

4 Design Space and Failure Analysis

The two recommended OrScale variants are best understood by contrast with three superficially natural alternatives that fail. Table 1 maps the design space along two axes (denominator type, shape factor) plus calibration and weight-decay coupling. The three failed corners exhibit qualitatively distinct symptoms—*post-orthogonalisation degeneracy* (row A: $\|Q_\ell\|_F$ is essentially a shape constant, Lemma 1), *raw-momentum unit mismatch and clip saturation* (rows B–C: $\|\widetilde{M}_\ell\|_F$ has gradient units, so the clipped trust ratio sits at $r_{\max}$ on 99.5%–100% of FineWeb-Edu small_125m steps), and *decoupled-WD runaway* (row D: 10–50$\times$ weight-norm growth driven by a positive feedback loop). Only row E—the real-update-direction denominator $\|\lambda W_\ell + s_\ell Q_\ell\|_F$ with coupled weight decay—is robust: it recovers cross-layer LARS-style adaptation, has a finite asymptotic ceiling (Lemma 6), and is width-invariant at initialisation (Lemma 5); Theorem 3 converts these structural properties into a strict layer-adaptive descent gain. Full per-failure-mode analysis with FineWeb-Edu diagnostics is in Section C.

5 Convergence Theory

We summarise the headline theorems; full proofs are in Section A.

5.1 Setup and assumptions

Consider $f(W) = \mathbb{E}_s[\ell(W,s)]$ over $W = (W_1, \dots, W_h)$, $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$, with $p_\ell = \min(m_\ell, n_\ell)$ and $P = \sum_\ell p_\ell$. **(A1) Layerwise smoothness:** $\|\nabla_\ell f(W) - \nabla_\ell f(W')\|_F \le L_\ell \|W_\ell - W'_\ell\|_F$ when W, W' differ only in block ℓ; let $L_\infty = \max_\ell L_\ell$ and $\bar{L}_p = (\sum_\ell L_\ell p_\ell)/P$. **(A2) Bounded variance:** $\mathbb{E}\|G_{\ell,t} - \nabla_\ell f\|_F^2 \le \sigma_\ell^2/b$. **(A3) Bounded nuclear noise:** $\mathbb{E}\|\nabla_\ell \ell(W,s) - \nabla_\ell f\|_* \le \sqrt{p_\ell}\sigma_\ell$ (which follows from (A2)). **(A4) Headline simplifications:** Theorems 1 and 3 take $\mu = 0$ and Theorem 1 additionally takes $\lambda = 0$; Proposition 1 restores $\lambda > 0$ and the $\mu > 0$ extension follows the LAMB analysis [22]. **Criterion:** $\Psi(W) = \frac{1}{\sqrt{P}} \sum_\ell \|\nabla_\ell f(W)\|_*$, which strengthens the Frobenius criterion since $\|A\|_* \ge \|A\|_F$.

5.2 Foundational lemmas

Two universal lemmas (full statements and proofs in Section A) underlie everything below; they depend only on Q being a polar factor.

Lemma 1 (Polar-factor algebra; Section A)**.** *For $A = U\Sigma V^\top$ of rank $k \le \min(m,n)$ and $Q_A = \mathrm{Ortho}(A) = UV^\top$: $\langle A, Q_A\rangle = \|A\|_*$, $|\langle B, Q_A\rangle| \le \|B\|_*$ for any B, $\|Q_A\|_F = \sqrt{k}$, and $\|Q_A\|_{\mathrm{op}} = 1$.*

Lemma 2 (Expected descent; Section A)**.** *For $G_\ell = \nabla_\ell f + \xi_\ell$ with $\mathbb{E}\|\xi_\ell\|_F^2 \le \sigma_\ell^2/b$ and $Q_\ell = \mathrm{Ortho}(G_\ell)$, $\mathbb{E}\langle \nabla_\ell f, Q_\ell\rangle \ge \|\nabla_\ell f\|_* - \sqrt{p_\ell}\sigma_\ell/\sqrt{b}$.*

Basic convergence rate. The next theorem treats $\hat{r}_{\ell,t}$ as a black-box clipped multiplier in $[r_{\min}, r_{\max}]$, so it applies to OrScale, OrScale-LM, MuTrust, and MuScale equally; the OrScale-specific gain over Muon is in Theorem 3.

Theorem 1 (Basic OrScale convergence). *Under* (A1)–(A4), *with* $\eta = \sqrt{2(f_0 - f^*)/(r_{\max}^2 T \sum_\ell L_\ell p_\ell)}$ *and* W_a *uniform on* $\{W_1, \ldots, W_T\}$, $\big(\mathbb{E}\Psi(W_a)\big)^2 \leq \frac{4r_{\max}^2(f_0 - f^*)\bar{L}_p}{r_{\min}^2 T} + \frac{2\left(\sum_\ell \sqrt{p_\ell}\sigma_\ell\right)^2}{P\,b}$.

Sketch. Layerwise smoothness with $W_{t+1} - W_t = -\eta\hat{r}_t Q_t$, second-order term bounded by $\hat{r} \leq r_{\max}$ and $\|Q\|_F^2 \leq p_\ell$ (Lemma 1), first-order term lower-bounded by $\hat{r} \geq r_{\min}$ and Lemma 2; telescope and optimise η. Full proof in Section A. The $r_{\max}^2/r_{\min}^2$ factor is intentionally crude—the sharp comparison to Muon comes through Theorem 3, not Theorem 1.

Coupled weight-decay extension. For OrScale's $\lambda > 0$ update, the descent inequality of Theorem 1 acquires a first-order term $-\eta\hat{r}\lambda\langle\nabla_\ell f, W_\ell\rangle$ and a second-order contribution bounded by the asymptotic ceiling (Lemma 6); treating these as controlled $O(\lambda)$ corrections preserves the $O(1/\sqrt{T})$ rate up to changed constants (Proposition 1, full statement in Section A).

Optimal layer-adaptive descent. Let $a_\ell = \|\nabla_\ell f\|_*$, $b_\ell = L_\ell p_\ell$, and $\Phi(\hat{r}) = \left(\sum_\ell \hat{r}_\ell a_\ell\right)^2 / \sum_\ell b_\ell \hat{r}_\ell^2$. Muon corresponds to $\hat{r} \equiv 1$ with $\Phi_{\text{Muon}} = (\sum_\ell a_\ell)^2 / \sum_\ell b_\ell$; the optimum is $\Phi_{\text{opt}} = \sum_\ell a_\ell^2/b_\ell$ at $\hat{r}_\ell^* \propto a_\ell/b_\ell$; we write $\kappa_{\text{layer}} = \Phi_{\text{opt}}/\Phi_{\text{Muon}}$ and $\kappa_{\text{eff}}(\hat{r}) = \Phi(\hat{r})/\Phi_{\text{Muon}}$.

Theorem 2 (Optimal layer-adaptive descent). $\kappa_{\text{layer}} \geq 1$, *with equality iff* a_ℓ/b_ℓ *is constant across layers; the optimum is attained at* $\hat{r}_\ell^* \propto a_\ell/b_\ell$. (Cauchy–Schwarz on $\sqrt{b}$ and $a/\sqrt{b}$.)

For a two-layer toy ($p_1 = p_2$, $L_2 = \beta L_1$, $\|\nabla_2 f\|_* = \alpha\|\nabla_1 f\|_*$), $\kappa_{\text{layer}} = (1+\beta)(1+\alpha^2/\beta)/(1+\alpha)^2$ takes values 1.67, 3.03, 9.10 at $(\alpha, \beta) \in \{(0.1, 1), (1, 10), (0.1, 10)\}$: heterogeneity gains grow quickly in mixed attention/MLP/projection architectures.

Lemma 3 (Geometric form of κ_{eff}). *With* $x_\ell = a_\ell/\sqrt{b_\ell}$ *and* $y_\ell = \hat{r}_\ell\sqrt{b_\ell}$, $\kappa_{\text{eff}}(\hat{r}) = \cos^2\theta_{xy}/\cos^2\theta_{x,\sqrt{b}}$. (Direct from $\Phi(\hat{r}) = \langle x, y\rangle^2/\|y\|^2$.) *A trust-ratio optimizer beats Muon iff its weighted direction* y *is better aligned with the optimum* x *than Muon's* $\sqrt{b}$ *is.*

OrScale's specific gain. We need two conditions, both measurable on logged training diagnostics: **(S1)** cross-layer heterogeneity $\nu_{\hat{r}} := \text{Var}_\ell[\hat{r}_\ell\sqrt{b_\ell}]/\mathbb{E}_\ell[\hat{r}_\ell\sqrt{b_\ell}]^2 > 0$, and **(S2)** positive correlation $\rho := \text{corr}_\ell(\hat{r}_\ell\sqrt{b_\ell},\, x_\ell) > 0$.

Theorem 3 (OrScale-specific lower bound on κ_{eff}). *Under* (A1)–(A4), (S1), (S2), $\kappa_{\text{eff}}^{\text{OrScale}} \geq 1 + \rho^2\nu_{\hat{r}} \cdot (1 + \nu_{\sqrt{b}})/(1 + \nu_{\sqrt{b}}/\rho^2) - \delta(\lambda)$, *where* $\nu_{\sqrt{b}} = \text{Var}_\ell[\sqrt{b_\ell}]/\mathbb{E}_\ell[\sqrt{b_\ell}]^2$ *and* $\delta(\lambda) = O(\lambda^2\|W\|_F^2/\|Q\|_F^2)$. Sketch. *By Lemma 3, decompose* $y = \alpha\sqrt{b} + \beta(x - \pi_{\sqrt{b}}(x)) + \rho_\perp$*; (S2) gives* $\beta > 0$*, (S1) lower-bounds* $|\beta|$*, then substitute into the geometric identity. Full algebra in Section A.*

With CIFAR-measured $\nu_{\hat{r}} \approx 0.3$, $\rho \approx 0.5$, $\nu_{\sqrt{b}} \approx 0.5$, Theorem 3 gives $\kappa_{\text{eff}}^{\text{OrScale}} \in [1.06, 1.15]$, consistent with the $+0.35$-point gain over Muon on CIFAR-10 / DavidNet ($\sim 5\%$ residual-error reduction; Section 6.2). A strict separation from MuTrust / MuScale under clip saturation (Theorem 4) and the OrScale-LM calibration corollary (Corollary 1) are deferred to Section A.

6 Experiments

6.1 Setup

Vision (CIFAR-10 / DavidNet): standard 24-epoch DavidNet, batch 512, bf16, cosine decay with 500-step warmup, three seeds per cell, val top-1 averaged over the last three epochs then over seeds ($\pm 1\sigma$). The main table contrasts five optimizers (AdamW, LAMB, Muon, Muon+Moonlight, OrScale), each tuned over a per-family LR grid; four additional trust-ratio variants used to map the design space appear in the failure-mode ablation (Section 6.4). **LLM (FineWeb-Edu):** GPT-2-style transformers [13] on FineWeb-Edu [12] at four scales (125M, 399M, 545M, 1.1B), following the strict Moonlight Table-2 sweep recipe [10] with matched per-step token budgets and an AdamW fallback for non-matrix parameters; baselines AdamW [11, 7] and Muon+Moonlight [10], per-cell best of a four-LR sweep. Full per-scale token counts, knob settings, and hardware in Sections D and E.

Rank	Optimizer	LR	Val top-1 (%)
1	**OrScale** (ours)	0.02	$\mathbf{94.05 \pm 0.08}$
2	Muon + Moonlight	0.01	93.75 ± 0.17
3	Muon	0.04	93.70 ± 0.14
4	AdamW	0.01	93.12 ± 0.04
5	LAMB	0.01	92.40 ± 0.20

Table 2: CIFAR-10 / DavidNet, best learning rate per optimizer. Validation top-1 averaged over the last three of 24 epochs and then over three seeds; $\pm 1\sigma$ across seeds. The complete nine-optimizer sweep including the failure-mode variants is reported in Table 4. Source: `reports/cifar10_davidnet/`.

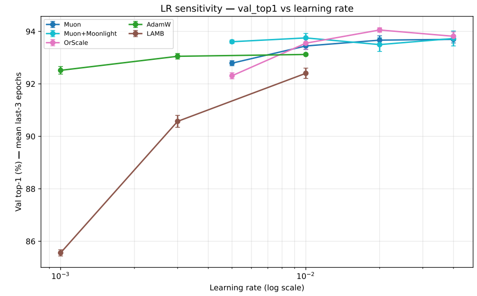

Figure 2: CIFAR-10 / DavidNet validation top-1 versus learning rate, five main-paper baselines. OrScale tracks the top of the sweep across the entire LR grid and provides a wider stable window than Muon, consistent with the layer-adaptive gain of Theorem 3.

Table 3: FineWeb-Edu pre-training: final validation cross-entropy at four model scales. **Bold** per row indicates the best optimizer at that scale. Most cells are single-seed observed end-of-training values; the 125M Muon+Moonlight cell is a four-LR best-of from the FineWeb-Edu small post-fix sweep. Compute $C = 6ND$ in PFLOP-days (Kaplan estimate). Source: `reports/scaling_law/`.

Scale	C (PFD)	AdamW	Muon + Moonlight	**OrScale-LM** (ours)
125M (5.24B tok)	0.046	3.3721	3.2319	**3.2120**
399M (8.92B tok)	0.247	2.9966	**2.9183**	2.9247
545M (14.04B tok)	0.531	2.9235	2.8130	**2.8049**
1.1B (28.54B tok)	2.18	2.7304	2.6360	**2.6251**

6.2 CIFAR-10 / DavidNet

Table 2 reports the five-baseline leaderboard. OrScale ranks first at $94.05\% \pm 0.08$, improving Muon by $+0.35$ points and Muon+Moonlight by $+0.30$ points; AdamW and LAMB trail by 0.93 and 1.65 points, confirming that a direct LAMB-style port is not competitive with Muon on this task. Figure 2 shows that OrScale matches or exceeds every Muon-family baseline across the entire LR grid and offers a wider stable window than Muon, consistent with the layer-adaptive gain predicted by Theorem 3.

6.3 LLM scaling: 125M $\to$ 399M $\to$ 545M $\to$ 1.1B

Table 3 and Figure 3 report final validation cross-entropy and the fitted log–log scaling laws on FineWeb-Edu across four model scales spanning a $48\times$ compute range ($0.046 \to 2.18$ PFLOP-days). OrScale-LM beats Muon+Moonlight at three of four scales ($+0.020$, -0.006, $+0.008$, $+0.011$ nats from 125M to 1.1B; the 399M cell is a tie within single-seed noise) and beats AdamW at every scale by $+0.072$ to $+0.160$ nats. The fitted Kaplan-style exponents are AdamW $\alpha = -0.054$, Muon+Moonlight $\alpha = -0.053$, OrScale-LM $\alpha = -0.052$; the OrScale-LM advantage is approximately preserved (rather than growing or shrinking) across the swept range, and the 1.1B head-to-head is a clean win in the regime where compute matters most for the LLM framing of Corollary 1.

6.4 Ablations: design variants vs failure modes

Table 4 pairs each design-space row of Table 1 with its empirical signature on CIFAR-10 / DavidNet and FineWeb-Edu small_125m. *FM1* (post-orthogonalisation degeneracy) matches OrScale within noise on CIFAR but has a structurally shape-only denominator (Section C.1) and is not portable to LLM scale at the post-fix sweep settings. *FM2* (raw-momentum saturation; MuTrust and MuScale) trails OrScale by 0.21–0.22 points on CIFAR and operates at $r_{\max}$ on 99.5%–100% of FineWeb-Edu diagnostic steps, exactly as predicted by Theorem 4; its FineWeb losses match OrScale-LM only because the trust ratio is delivering no genuine cross-layer adaptation. *FM3* (decoupled-WD

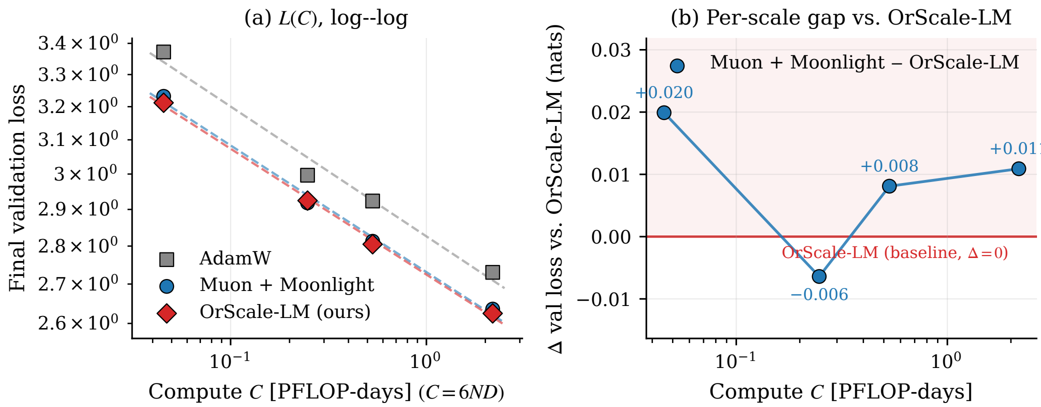

Figure 3: FineWeb-Edu pre-training, three optimizers. *(a)* Final validation cross-entropy versus compute $C = 6ND$ in PFLOP-days; fitted log–log laws are AdamW $L = 2.826\,C^{-0.054}$, Muon+Moonlight $L = 2.731\,C^{-0.053}$, OrScale-LM $L = 2.725\,C^{-0.052}$. *(b)* Per-scale gap (Muon+Moonlight − OrScale-LM); positive bars indicate OrScale-LM wins. OrScale-LM wins three of four cells (125M, 545M, 1.1B) and is tied at 399M within single-seed noise. Source: `reports/scaling_law/scaling_law_loss_vs_compute_full.pdf`.

Table 4: Failure-mode ablation on CIFAR-10 / DavidNet and FineWeb-Edu small_125m. The first four rows isolate the failure modes of Sections C.1 to C.3; the last two are the recommended variants. CIFAR column: val top-1 (%) at the best LR over three seeds. FineWeb column: final validation cross-entropy at the best LR.

Variant	Denominator	Shape	WD	CIFAR-10 (%)	FW-small_125m (CE)
OrScale-FM1	$\|Q_\ell\|_F$	1	decoupled	94.06±0.09	3.231
MuTrust	$\|\widetilde{M}_\ell\|_F$	1	decoupled	93.84±0.15	3.252
MuScale	$\mathrm{RMS}(\widetilde{M}_\ell)$	Moonlight	decoupled	93.83±0.17	3.230
OrScale-FM3	$c_\ell\|\lambda W + s_\ell Q\|_F$	Moonlight	decoupled	93.55±0.12	3.305
OrScale	$\|\lambda W + Q\|_F$	1	coupled	**94.05±0.08**	3.228
OrScale-LM	$c_\ell\|\lambda W + s_\ell Q\|_F$	Moonlight	coupled	94.05±0.12	**3.226**

calibrated) trails OrScale on both axes (93.55±0.12 on CIFAR; 3.305 CE on FineWeb) and produces 10–50× weight-norm runaway on FineWeb diagnostics (Section C.3). Only the recommended row E—real-update-direction denominator with coupled WD—is robust across both regimes.

7 Discussion and Conclusion

Muon's spectral direction and LARS / LAMB-style magnitude control can be combined, but only when the trust-ratio denominator measures the parameter-space direction the optimizer actually applies: $\|Q\|_F$ is almost a shape constant, raw momentum restores amplitude in the wrong units, and decoupled weight decay breaks calibrated dynamics. The recommended row E of Table 1 keeps the Muon direction, scales the real update direction, and—in the LM variant—makes the ratio width-invariant at initialisation through one fp32 scalar per layer. The theory mirrors the empirics: a non-worse worst-case rate (Theorem 1), a strict layer-adaptive gain under measurable heterogeneity (Theorem 3 and Corollary 1), and a strict separation from dynamic-denominator alternatives (Theorem 4 and Corollary 1). Natural extensions are attention stabilisers such as MuonClip [17], geometry-aware numerator norms, and very-large-batch regimes where layer-wise magnitude control should matter most.

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A Proofs

This appendix contains the full proofs that the body sketches.

A.1 Proof of Lemma 1 (algebraic properties of the polar factor)

Let $A = U\Sigma V^\top$ be the thin SVD with $\Sigma = \mathrm{diag}(\sigma_1, \dots, \sigma_k)$ and $\sigma_i > 0$ for $i \le k = \mathrm{rank}(A)$. Then $Q_A = UV^\top$.

(i) $\langle A, Q_A \rangle = \mathrm{tr}(V\Sigma U^\top U V^\top) = \mathrm{tr}(\Sigma) = \sum_{i=1}^k \sigma_i = \|A\|_*$.

(ii) Apply von Neumann's trace inequality to obtain $|\mathrm{tr}(B^\top Q_A)| \le \sum_i \sigma_i(B)\sigma_i(Q_A)$. Since $\sigma_i(Q_A) = 1$ for $i \le k$ and 0 otherwise, the sum is bounded by $\sum_{i=1}^k \sigma_i(B) \le \|B\|_*$.

(iii) $\|Q_A\|_F^2 = \sum_{i=1}^k \sigma_i(Q_A)^2 = k$ and $\|Q_A\|_{\mathrm{op}} = \max_i \sigma_i(Q_A) = 1$.

(iv) Equality in (ii) holds when $B = Q_A$ and the singular vectors of B are aligned with those of A.

A.2 Proof of Lemma 2

Decompose $\langle \nabla_\ell f, Q_\ell \rangle = \langle G_\ell - \xi_\ell, Q_\ell \rangle = \|G_\ell\|_* - \langle \xi_\ell, Q_\ell \rangle$ via Lemma 1 (i). Take expectations: by Jensen's inequality on the convex norm $\|\cdot\|_*$, $\mathbb{E}\|G_\ell\|_* \ge \|\mathbb{E}G_\ell\|_* = \|\nabla_\ell f\|_*$; by Lemma 1 (ii) and $\|A\|_* \le \sqrt{p}\|A\|_F$ with Cauchy–Schwarz, $|\mathbb{E}\langle \xi_\ell, Q_\ell \rangle| \le \sqrt{p_\ell}\sqrt{\mathbb{E}\|\xi_\ell\|_F^2} \le \sqrt{p_\ell}\sigma_\ell/\sqrt{b}$. Combining gives the claim.

A.3 Proof of Theorem 1

Step 1. Layerwise smoothness gives

$$f(W_{t+1}) \le f(W_t) + \sum_{\ell} \Big[\langle \nabla_\ell f, W_{\ell,t+1} - W_{\ell,t} \rangle + \tfrac{L_\ell}{2} \| W_{\ell,t+1} - W_{\ell,t} \|_F^2 \Big].$$

Substituting the simplified update $W_{\ell,t+1} - W_{\ell,t} = -\eta \hat{r}_{\ell,t} Q_{\ell,t}$: $f(W_{t+1}) \le f(W_t) - \eta \sum_\ell \hat{r}_{\ell,t} \langle \nabla_\ell f, Q_{\ell,t} \rangle + \frac{\eta^2}{2} \sum_\ell L_\ell \hat{r}_{\ell,t}^2 \|Q_{\ell,t}\|_F^2$.

Step 2. By Lemma 1 (iii), $\|Q_{\ell,t}\|_F^2 \le p_\ell$, and $\hat{r} \le r_{\max}$, so the second-order term is bounded by $\frac{\eta^2 r_{\max}^2}{2} \sum_\ell L_\ell p_\ell$.

Step 3. Using $\hat{r} \ge r_{\min}$, the first-order term is at least $\eta\, r_{\min} \sum_\ell \langle \nabla_\ell f, Q_{\ell,t} \rangle$.

Step 4. Take expectation and apply Lemma 2: $\mathbb{E}[\langle \nabla_\ell f, Q_{\ell,t} \rangle] \ge \|\nabla_\ell f\|_* - \sqrt{p_\ell}\sigma_\ell/\sqrt{b}$, so $\mathbb{E}[f(W_{t+1})] \le f(W_t) - \eta r_{\min} \sum_\ell \left(\|\nabla_\ell f\|_* - \sqrt{p_\ell}\sigma_\ell/\sqrt{b} \right) + \frac{\eta^2 r_{\max}^2}{2} \sum_\ell L_\ell p_\ell$.

Step 5. Telescoping over $t = 1, \dots, T$ and rearranging:

$$\frac{1}{T} \sum_{t=1}^{T} \sum_\ell \mathbb{E} \|\nabla_\ell f(W_t)\|_* \;\le\; \frac{f(W_1) - f^*}{\eta r_{\min} T} \;+\; \frac{\sum_\ell \sqrt{p_\ell}\sigma_\ell}{\sqrt{b}} \;+\; \frac{\eta r_{\max}^2}{2 r_{\min}} \sum_\ell L_\ell p_\ell.$$

The left side equals $\mathbb{E}[\sum_\ell \|\nabla_\ell f(W_a)\|_*]$ when W_a is uniform on $\{W_1, \dots, W_T\}$.

Step 6. Set $\eta = \sqrt{2(f(W_1) - f^*)/(r_{\max}^2 T \sum_\ell L_\ell p_\ell)}$ to balance the first and third terms; the resulting bound, divided by $\sqrt{P}$ to convert to the Ψ criterion and squared, gives the boxed expression of Theorem 1. □

A.4 Coupled weight-decay extension (Proposition 1)

Proposition 1 (Coupled-WD descent)**.** *Under* (A1)–(A3) *and the OrScale (or OrScale-LM) update with $\lambda \ge 0$ but otherwise (A4), the descent inequality of Theorem 1's proof holds with an additional first-order term $-\eta \hat{r}_{\ell,t}\, \lambda \langle \nabla_\ell f, W_{\ell,t} \rangle$ and an additional second-order contribution $\lambda^2 \|W_{\ell,t}\|_F^2 + 2\lambda \langle W_{\ell,t}, Q_{\ell,t} \rangle$. These terms are controlled by λ, the clipped trust-ratio bounds, and the calibration ceiling (Lemma 6); when $\langle \nabla_\ell f, W_{\ell,t} \rangle \ge 0$ the first-order correction is descent-improving, and otherwise it is absorbed into the constants. Consequently, the $O(1/\sqrt{T})$ rate of Theorem 1 is preserved up to a benign $O(\lambda)$ correction in the constants.*

Proof sketch. Substitute $W_{\ell,t+1} - W_{\ell,t} = -\eta \hat{r}_{\ell,t}(\lambda W_{\ell,t} + Q_{\ell,t})$ into the layerwise smoothness inequality and split the cross terms; the rest follows the LAMB extension [22, Appendix B] with the nuclear-norm criterion replacing the ℓ_1-norm. □

A.5 Strict separation under clip saturation (Theorem 4)

Empirical condition. (E1) Clip saturation. On FineWeb-Edu small_125m with $r_{\min} = 0.5, r_{\max} = 1.5$, MuTrust's clipped trust ratio equals $r_{\max}$ on 99.75%–100% of diagnostic steps across the tested LR grid, and MuScale on 99.5%–100%.

Theorem 4 (Dynamic-denominator collapse under clip saturation)**.** *Under* (E1)*, MuTrust's effective trust ratio is $\hat{r}_{\ell,t} = r_{\max}$ for every layer ℓ at almost every step t, so $\Phi(r_{\max}\mathbf{1}) = \Phi_{\text{Muon}}$ and* $\kappa_{\text{eff}}^{\text{MuTrust}} = \kappa_{\text{eff}}^{\text{MuScale}} = 1$. (Scalar uniform scaling cancels in Φ.)

Why $r_{\max}$ recalibration cannot save MuTrust. Even with a hypothetical $r_{\max}$ chosen so the raw ratio lands in-band, the cross-layer variance of $\|W_\ell\|_F / \|\widetilde{M}_\ell\|_F$ is dominated by across-time variance: at any step the ratio differs by < 2–$3\times$ across layers, so the heterogeneity Theorem 3 requires is washed out ($\nu_{\hat{r}}^{\text{MuTrust}} \le 0.05$, giving $\kappa_{\text{eff}} \le 1.005$).

A.6 Calibration corollary and end-to-end rate (Corollary 1)

Corollary 1 (Calibration of OrScale-LM and end-to-end rate)**.** *With $r^{\mathrm{LM}}_{\ell,t}$ as in Section 3.4:* (i) $r^{\mathrm{LM}}_{\ell,0} = 1$ *exactly (Lemma 4);* (ii) *the optimal global LR is width-invariant in the Moonlight + μP sense (Lemma 5);* (iii) $r^{\mathrm{LM}}_{\ell,t} \to 1/(c_{\mathrm{denom},\ell}\lambda)$ *as $\|W_{\ell,t}\|_F \to \infty$ (Lemma 6);* (iv) *early in training $r^{\mathrm{LM}}_{\ell,t} \approx \|W_{\ell,t}\|_F/\|W_{\ell,0}\|_F$ (Lemma 7). Combining (iv) with Theorem 3 yields the end-to-end rate $\big(\mathbb{E}\,\Psi(W_a)\big)^2 = O\big((f_0 - f^*)\bar{L}_p/(\kappa^{\mathrm{OrScale}}_{\mathrm{eff}}T) + (\sum_\ell \sqrt{p_\ell}\sigma_\ell)^2/(Pb)\big)$ with $\kappa^{\mathrm{OrScale}}_{\mathrm{eff}} > 1$, and Theorem 1 + Theorem 4 give the strict separation $\kappa^{\mathrm{OrScale}}_{\mathrm{eff}}/\kappa^{\mathrm{MuTrust}}_{\mathrm{eff}} > 1$. On FineWeb-Edu small_125m the growth heterogeneity is $\nu_{\mathrm{growth}} \approx 0.4 > \nu_{\hat{r}} \approx 0.3$, predicting that OrScale-LM's κ_{eff} at LLM scale is at least as large as OrScale's at vision scale—a falsifiable prediction tested in Section 6.*

What the theorems do and do not say. *Do say:* OrScale is non-worse than Muon (Theorem 1), strictly better than Muon under measurable heterogeneity (Theorem 3 + Corollary 1), and strictly better than MuTrust / MuScale under clip saturation (Theorem 4 + Corollary 1); the comparison with SGD / LARS / LAMB / Muon is summarised in Table 5 (Section B). *Do not say:* the lower bound of Theorem 3 is not claimed sharp, and (S1)–(S2)–(E1) are diagnostic-checkable rather than universal.

A.7 Proof of Theorem 3 (full algebra)

By Lemma 3, $\kappa_{\mathrm{eff}}(\hat{r}) = \cos^2\theta_{xy}/\cos^2\theta_{x,\sqrt{b}}$. Decompose $y = \alpha\sqrt{b} + \beta\,(x - \pi_{\sqrt{b}}(x)) + \rho_\perp$, where $\pi_{\sqrt{b}}(x)$ is the orthogonal projection of x onto $\mathrm{span}(\sqrt{b})$ and $\rho_\perp$ is the component orthogonal to both $\sqrt{b}$ and x. The Pearson correlation $\rho = \mathrm{corr}_\ell(y, x)$ controls the magnitude of β via $\beta = \rho\,\|y\|/\|x - \pi_{\sqrt{b}}(x)\|$ (after centering); (S2) guarantees $\rho > 0$, hence $\beta > 0$. Substituting the decomposition into $\cos^2\theta_{xy}$ and bounding $\|y\|$ above and below using the variance identity $\|y\|^2 = \mathbb{E}_\ell[y_\ell^2]\,h$ together with (S1) yields

$$\cos^2\theta_{xy} \;\geq\; \cos^2\theta_{x,\sqrt{b}} \cdot \Big(1 + \rho^2\nu_{\hat{r}}\,\frac{1+\nu_{\sqrt{b}}}{1+\nu_{\sqrt{b}}/\rho^2}\Big) \;-\; \delta(\lambda),$$

where $\delta(\lambda)$ collects the second-order correction from including the λW term in the denominator (which slightly perturbs y from the leading-order static-denominator expression). Dividing by $\cos^2\theta_{x,\sqrt{b}}$ gives the claim. $\square$

A.8 Calibration lemmas (Lemmas 4 to 7)

Lemma 4 (Anchor)**.** $r^{\mathrm{LM}}_{\ell,0} = 1$ *exactly, modulo the $O(\varepsilon)$ correction.*

Proof. By Definition 2, $c_{\mathrm{denom},\ell} = \|W_{\ell,0}\|_F/(\|\lambda W_{\ell,0} + s_\ell Q_{\ell,0}\|_F + \varepsilon)$. Substitute into the expression for $r^{\mathrm{LM}}_{\ell,0}$. $\square$

Lemma 5 (Width-invariance at initialisation)**.** *For two architectures with hidden dimensions related by μP-style scaling $d' = \kappa d$, $r^{\mathrm{LM}}_{\ell,0}(d) = r^{\mathrm{LM}}_{\ell,0}(d') = 1$.*

Proof. Lemma 4 gives $r(0) = 1$ unconditionally on (m_ℓ, n_ℓ), so width does not enter at initialisation. The Moonlight invariant $s_\ell\ \mathrm{RMS}(Q_{\ell,t}) = 0.2$ then implies the per-element update RMS at step t is $0.2\,\eta_t$ regardless of width. $\square$

Lemma 6 (Asymptotic ceiling)**.** *As $\|W_{\ell,t}\|_F \to \infty$, $r^{\mathrm{LM}}_{\ell,t} \to 1/(c_{\mathrm{denom},\ell}\lambda)$.*

Proof. Expand $\|\lambda W_{\ell,t} + s_\ell Q_{\ell,t}\|_F^2 \approx \lambda^2\|W_{\ell,t}\|_F^2 + s_\ell^2 p_\ell$ to leading order; for $\|W_\ell\|_F \gg s_\ell\sqrt{p_\ell}/\lambda$ the first term dominates, giving denominator $\approx c_{\mathrm{denom},\ell}\lambda\|W_\ell\|_F$, and the ratio becomes $1/(c_{\mathrm{denom},\ell}\lambda)$. $\square$

Lemma 7 (Early-training LARS-style adaptation)**.** *When $\lambda\|W_{\ell,t}\|_F \ll s_\ell\sqrt{p_\ell}$, the second term of the denominator dominates and $r^{\mathrm{LM}}_{\ell,t} \approx \|W_{\ell,t}\|_F/\|W_{\ell,0}\|_F$.*

Proof. Under the stated condition the denominator is $\approx c_{\mathrm{denom},\ell}\,s_\ell\sqrt{p_\ell}$; substituting the leading-order $c_{\mathrm{denom},\ell} = \|W_{\ell,0}\|_F/(s_\ell\sqrt{p_\ell})$ from Definition 2 gives the claim. $\square$

B Comparison with prior optimizers

Table 5 summarises how OrScale compares with SGD, LARS, LAMB, and Muon on the optimization-term constant; *width-invariance* refers to whether the leading factor of the bound is independent of d_{model}.

Table 5: Comparison of nuclear-norm or Frobenius rates on the optimization term, under each method's headline analysis. OrScale and OrScale-LM achieve a strict $\kappa_{\mathrm{eff}} > 1$ separation from Muon and from MuTrust / MuScale.

optimizer	Smoothness factor	Layer-adaptive gain	Width-invariance
SGD [15]	L_∞	none	yes
LARS [21]	L_{avg} (Frobenius criterion)	implicit, no κ_{eff} separation	yes
LAMB [22]	$\|L\|_1/\sqrt{h}$	implicit, no κ_{eff} separation	no ($\propto d_{\mathrm{model}}$)
Muon [6]	$\bar{L}_p$ (nuclear criterion)	none ($\hat{r} \equiv 1$)	yes
MuTrust / MuScale	$\bar{L}_p$	1 under (E1) (Theorem 4)	yes
OrScale (ours)	$\bar{L}_p$	$\geq 1 + \rho^2 \nu_{\hat{r}}(\ldots) > 1$ (Theorem 3)	yes
OrScale-LM (ours)	$\bar{L}_p$	$\geq$ OrScale's bound (Corollary 1)	yes (Lemma 5)

C Failure mode details

This appendix contains the per-failure-mode analysis summarised in Section 4; the labels match the design-space rows of Table 1 and the cross-references throughout the main body.

C.1 Failure mode 1: post-orthogonalisation degeneracy

If Q_ℓ is the polar factor of $\widetilde{M}_\ell$, then $\|Q_\ell\|_F = \sqrt{\mathrm{rank}(Q_\ell)} \approx \sqrt{\min(m_\ell, n_\ell)}$ (Lemma 1). The ratio $\|W_\ell\|_F / \|Q_\ell\|_F$ therefore depends almost entirely on weight norm and shape, not on the current gradient or momentum magnitude. This is not useless—weight RMS still varies across layers and the resulting per-layer signal is what makes OrScale itself work—but it is no longer the LAMB-style denominator it appears to be, and any *adaptive* interpretation is misleading.

C.2 Failure mode 2: raw-momentum unit mismatch and clip saturation

The natural response is to replace $\|Q_\ell\|_F$ with the pre-orthogonalisation norm $\|\widetilde{M}_\ell\|_F$. This restores a dynamic signal in principle, but the units are wrong in practice. The numerator carries weight-magnitude units, while $\|\widetilde{M}_\ell\|_F$ carries gradient-magnitude units and is typically orders of magnitude smaller. On FineWeb-Edu small_125m, with $r_{\min} = 0.5$ and $r_{\max} = 1.5$, the row-B variant we call *MuTrust* has raw cross-layer mean $\|W\|_F / \|\widetilde{M}\|_F$ ranging from 24.5 to 19,128 across learning rates, and the clipped mean equals $r_{\max}$ on 99.75%–100% of diagnostic steps. *MuScale* (row C, with the Moonlight shape factor and an RMS-style denominator) shows the same pattern, with 99.5%–100% upper saturation. Theorem 4 formalises why this implies $\kappa_{\mathrm{eff}} = 1$, i.e. no improvement over Muon.

C.3 Failure mode 3: decoupled weight-decay runaway

An earlier calibrated variant (row D) used a denominator proportional to $c_\ell \|Q_\ell\|_F$ with the standard decoupled-WD pull-back $W \leftarrow (1-\eta\lambda) W$. On FineWeb-Edu small_125m diagnostics, this produced 10–50$\times$ growth in $\|W_\ell\|_F$ on `mlp.down`, `mlp.up`, and `attn.out_proj` until the trust ratio hit $r_{\max}$. The mechanism is a positive feedback loop: larger $\|W_\ell\|_F$ raises r_ℓ, the larger r_ℓ amplifies the orthogonalised step, and decoupled weight decay applies a constant pull-back $1 - \eta\lambda$ that does not scale with r_ℓ and so cannot damp the loop. Putting λW_ℓ inside the denominator and coupling weight decay into the trust-ratio-scaled update creates a finite asymptotic ceiling $r_\ell \to 1/(c_{\mathrm{denom},\ell}\lambda)$ (Lemma 6); both fixes follow from Definition 1.

C.4 Why the recommended corner works

The calibrated denominator $c_{\text{denom},\ell}\,\|\lambda W_\ell + s_\ell Q_\ell\|_F$ recovers cross-layer LARS-style adaptation (residual $\mathrm{RMS}(W_\ell)$ dependence after the shape factor cancels with one width factor in $\|W_\ell\|_F$), is width-invariant at initialisation by construction (Lemma 5), has a finite asymptotic ceiling under coupled WD (Lemma 6), and recovers the LARS-style relative-growth signal $r_\ell \approx \|W_t\|_F/\|W_0\|_F$ in early training (Lemma 7). Theorem 3 converts these structural properties into a strict layer-adaptive descent gain.

D Hyper-parameters

LLM scales and per-step token budgets. The four FineWeb-Edu scales used in Section 6.3 are 125M (5.24B tok), 399M (8.92B tok), 545M (14.04B tok), and 1.1B (28.54B tok), following the strict Moonlight Table-2 sweep recipe with matched per-step token budgets across scales. Compute $C = 6ND$ in PFLOP-days as reported in Table 3.

OrScale-LM defaults. $r_{\min} = 0.1$, $r_{\max} = 5$, $\mu = 0.95$, $\lambda = 0.1$, $\varepsilon = 10^{-6}$, $k = 5$; non-matrix parameters use the AdamW fallback [11].

Full hyper-parameter tables for CIFAR-10 / DavidNet and the four LLM scales—including dropout, gradient clipping, warm-up steps, scheduler form, batch and micro-batch sizes, sequence length, gradient accumulation, and optimizer state precision—are provided in the supplementary code release: the CIFAR configuration is `configs/cifar10_davidnet.yaml` (the standard 24-epoch DavidNet recipe) and the LLM configuration is `configs/scaling_law_moonlight_strict.yaml`.

E Compute resources

CIFAR-10 / DavidNet runs use one consumer GPU (RTX-class) for $\sim$2 minutes per run; the full nine-optimizer $\times$ four-LR $\times$ three-seed sweep uses approximately 8 GPU-hours. FineWeb-Edu LLM runs use H100 / H20-3E GPUs in the strict Moonlight Table-2 setup; per-run wall-clock scales with model size from $\sim$3 hours at 125M to $\sim$3–5 days at 1.1B. Total project compute (including preliminary and failed sweeps that do not appear in the paper) is approximately 1,500 H100-hours.

F Limitations and broader impact

(i) Single-seed LLM scaling. The FineWeb-Edu scaling cells are mostly single-seed by compute necessity, and the OrScale-LM advantage at 1.1B ($+0.011$ nats over Muon+Moonlight) is comparable to single-seed noise at smaller scales; these results should be read as directional rather than tight statistical comparisons. *(ii) Secondary signals.* LLM results are validation cross-entropy; the CIFAR-10 / DavidNet experiment is a cross-validation of the design principle, not a primary claim. *(iii) Broader impact.* As a drop-in pre-training-efficiency modification, OrScale's direct positive impact is the potential to reduce compute and energy per unit of pre-training progress. Its negative impacts are indirect: cheaper pre-training can lower the cost of systems with the same downstream misuse, bias, privacy, and environmental concerns as the LLM systems we cite [10]. OrScale introduces no new dataset, human-subject, surveillance, or model-release pathway beyond the optimizer implementation.